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EX NO: 9	Massive MIMO Capacity vs SNR and Gain Using MATLAB
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Aim

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

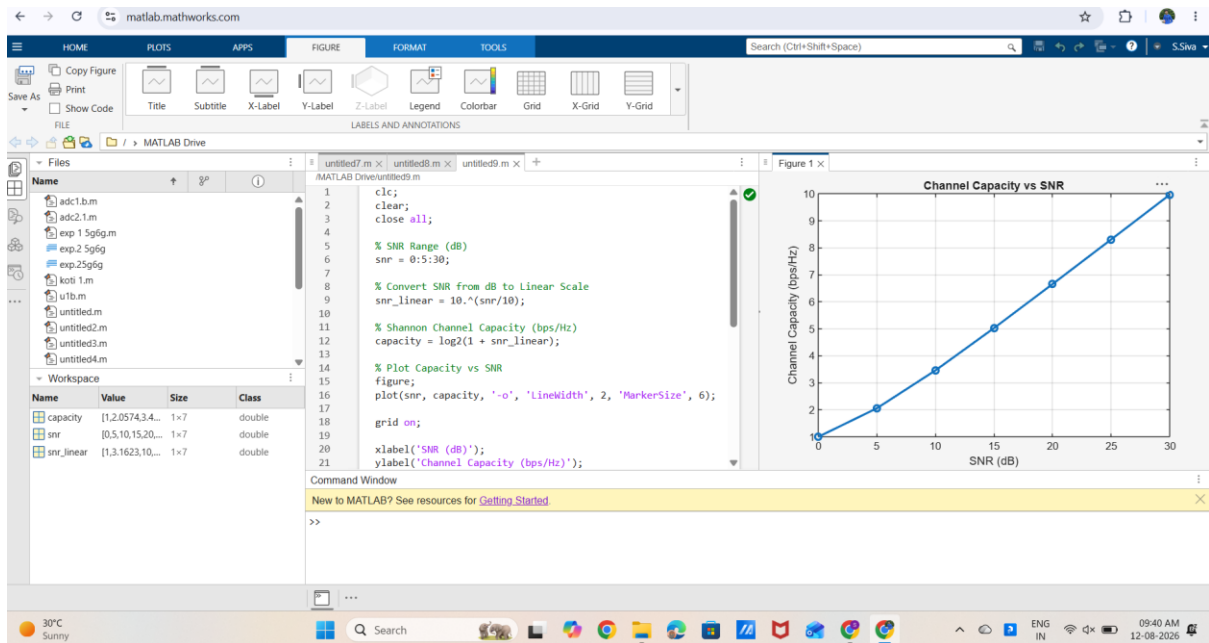
Software Required

- MATLAB

MATLAB Code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.